Kevin Prashan

Roll No.: CH23B019
B.Tech - Chemical Engineering
Indian Institute Of Technology, Tirupati

+91-8122663608
pkevinprashan@gmail.com
ch23b019@iittp.ac.in
LinkedIn

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (CHE)	Indian Institute of Technology Tirupati	7.35	2023-2027
Senior Secondary	Central Board of Secondary Education	91.6%	2023
Secondary	Central Board of Secondary Education	77.2%	2021

SUMMARY

I am a Chemical Engineering student at IIT Tirupati with a keen interest in **Process Control** and **Reaction Engineering**. I have gained hands-on experience with **ASPEN Plus** and **MATLAB** through academic projects, such as modeling ammonia synthesis processes. I enjoy applying theoretical concepts to practical problems, a mindset which i also apply in cooking, an approach that helped me secure 1st place in a Simulink pH control hackathon and 1st place in mixology at Inter IIT 8.0.

Projects

- **UHDE Ammonia Synthesis** *August 2024 - November 2025*
  *Academic Project/ **Guide**: Dr.Nilesh, Department of Chemical Engineering*
  - **Tools & technologies used**: Microsoft Excel, ASPEN Plus
  - Collaborated with a team of 5 and Developed a comprehensive Process Flow Diagram (PFD) and performed material and energy balances for a large-scale UHDE Ammonia Synthesis process.
  - Modeled and optimized the UHDE process, performing energy balance calculations for a 1700 MTPD plant
  - Analyzed pressure and temperature drops across 3 reactor beds, 2 Separators and Mixers, recommending changes to maintain a target conversion rate of 98%
  - Performed Adsorption and Absorption analysis on ammonia synthesis loop and CO2 removal to find the reactor size and amount of adsorbant required for the plant
  - Studied about implying control based on extended stoltze's model, accounting for hydrogen inhibition at temperatures lower than 623k
- **Viscous and Inertial Propulsion:** *Jan. 2025 - April 2025*
  *Academic Project/ **Guide**: Prof. Anki, Department of Chemical Engineering*
  - Designed and demonstrated propulsion in fluids through viscous and inertial effects using oscillatory and reciprocal motion of symmetric bodies.
  - Analyzed two distinct propulsion regimes: the viscous regime (low Reynolds number) and the inertial regime.
  - Experimentally validated that asymmetrical stroke patterns in Newtonian fluids produce net displacement without external force.
  - Studied applications in bio-inspired robotics and microfluidic devices where traditional propulsion is ineffective.
- **Estimation of solar radiation over sloped topography** *Sept. 2024 – Nov. 2024*
  *Academic Project | **Guide**: Dr. Panchatcharam Mariappan, Department of Mathematics and Statistics*
  - Analyzed the solar energy potential of mountainous terrain by processing latitude, longitude, and altitude across 2900 sample points.
  - Implemented the Iqbal model to simulate direct, diffuse, and reflected solar radiation, incorporating complex variables such as slope correction factors and sky view factors.
  - Calculated a total daily solar radiation of 6391.19 Wh/m^2 by identifying peak irradiance and optimal slope angles up to 40 degrees.
- **Stirrer speed setpoint tracking with PID tuning** *Oct. 2025 - Nov. 2025*
  *Academic Project | **Guide**: Dr. Nabil M, Department of Chemical Engineering*
  - Successfully demonstrated a working model of stirrer which was able to track the given setpoint and reject any external the disturbance
  - It works on the principle by measuring back EMF generated by the stirrer for a given speed.
- **Viscosity Control for a Non-Newtonian Fluid Without Viscometer(Discontinued)** *Aug. 2025 - Oct. 2025*
  *Academic Project | **Guide**: Dr. Nabil M, Department of Chemical Engineering*
  - Tried demonstrating viscosity control without a viscometer using back EMF generated by stirrer as manipulated variable.
  - The heat generated by 12V heating coil was not sufficient to change the viscosity significantly for a Non-Newtonian fluid, thus the setpoint tracking was hard

- **ALLORAH: Awareness website** *Jan. 2025 - May 2025*
  *Academic Project| **Guide:** Dr.Suresh, Department of Civil and Environmental Engineering*
  - **Tools & technologies used**: HTML, CSS, javascript, three.js, Blender
  - Led a team of 15 students to design and launch an interactive website, presenting the project to an audience of 240+ faculty and students.
  - Developed a 3D explorable map of IIT Tirupati, integrated 6 mini-games, and co-created 3D animated videos from scratch.

Technical Skills

–**Programming:** C/C++, Python, Matlab

–**Tools & OS:** ASPEN Plus, Autocad, Blender, MS-Office, Linux, Windows

–**Libraries/Frameworks:** Pandas, Numpy, scikit-learn, dotenv, Selenium Webdriver

Positions of Responsibility

- **General Manager, E-Cell**
  Indian Institute of Technology, Tirupati July 2025 – Present
  * (Promoted from Executive) As Executive, managed content strategy and branding for flagship events, increasing social media engagement by over 30% through targeted campaigns.
  * Oversee strategy and operations along side the head, for 7 verticals with a total team size of approximately 60 members.
  * Helped managing workshop on stock market trading and E-Auction, which had 100+ participants each.
  * Hosted a workshop and quiz on "Steps to become an Entrepreneur" which had 50+ participants.
- **Cell Lead of Finance Cell, FMC Nexus**
  Indian Institute of Technology, Tirupati August 2025 – Present
  * The Finance, Marketing and Consulting club of IIT Tirupati, as a founding Finance Cell lead, I seek for collaborations with investment banking companies and related companies to work on live projects related to the field.
  * Hosted an event to simulate trading and stock market in a classroom, which had about 30+ participation.
  * Worked alongside consultancy cell to analyze why GROMO's new customers on their app leave early and presented it to a representative from GROMO.
- **Executive, TEDx IIT Tirupati – Production and Planning Team**
  Indian Institute of Technology, Tirupati Nov 2024 – Feb 2025
  * Directed planning within budget constraints, delivering thematic designs for the very first TEDX IIT T .
  * Assisted with logistics, venue setup, and speaker coordination during the event execution.
- **Core Member, Tirutsava – Techno-Cultural Fest (Marketing and Outreach Committee)**
  Indian Institute of Technology, Tirupati Sept 2024 – Feb 2025
  * Managed external communications and outreach campaigns for 10 institutions to boost student participation from other institutions.
  * Analyzed over 100+ music artists manager's contacts, and finalizing the artist based on budget constraints and trending artist.

Certifications

- Certification in Completion of Industrial Automation in pneumatics and hydraulics. **SMC Coorporation, Japan**

Achievements

* Won 1st in 24 hour simulink hackathon on pH control . **ChEC IIT Tirupati**
* Secured 2nd in Poster presentation on Ammonia synthesis. **Alchemia 1.0**
* Won 1st in mixology . **Inter IIT 8.0**
* Won 1st in chef clash . **Intra IIT 7.0**